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PERFORMANCE SPECIFICATION

AUGER, EARTH, TRUCK-MOUNTED, 91 CENTIMETER (36 INCH) DIAMETER HOLE, 18 METER (60 FOOT) MAX DEPTH (METRIC)

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers a diesel-engine-driven auger for digging large diameter holes and setting casings.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This specification does not include documents cited in other sections of this specification or recommended for addition information or as examples. While every effort has been made to ensure the completeness of the list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addresssed to : US Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 3820

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

FEDERAL

Code of Federal Regulations (CFR).

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

DEPARTMENT OF DEFENSE

7056674	- Cable, Waterproof, Electrical.
8343583	- Cradle, Chock Block.
8343584	- Block Assy., Chock.

(Copies of this drawing are available from the U.S. Army Tank-automotive and Armaments Command, AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ASQC Z1.4 - Sampling Procedures and Tables for Inspections by Attributes (DoD Adopted).

(Application for copies of ASQC Standards may be obtained from the American Society for Quality Control, P.O. Box 305, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D380 -Rubber Hose.

(Application for copies of ASTM Standards may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

STATE OF CALIFORNIA DEPARTMENT OF MOTOR VEHICLES

California Motor Vehicles Code.

(Application for copies should be addressed to the State of California, Department of Motor Vehicles, P.O. Box 1319, Sacramento, CA. 95812)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

J383	- Motor Vehicle Seat Belt Anchorages - Design Recommendations, Recommended Practice.
J636	- V-Belts and Pulleys, Standard.
J898	- Control Locations for Off-Road Self-Propelled Work Machines,Standard.
J899	- Operator’s Seat Dimensions for Off-Road Self-Propelled Work Machines, Standard.
J1834	- Seat Belt Comfort, Fit, and Convenience, Recommended Practice.

(Application for copies should be addressed to Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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TIRE AND RIM ASSOCIATION, INCORPORATION (T&RA)

Tire and Rim Yearbook.

(Applications for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Suite 304, Akron, OH 44313.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Unless otherwise specified herein or specified in the applicable drawings, materials used shall be in accordance with the manufacturer's material specifications for augers. The materials shall be capable of meeting all of the operational and environmental requirements specified herein (see 4.5.1). Recovered materials shall be used to the maximum extent practicable. Materials and special treatments shall be utilized to prevent corrosion and to resist harmful effects of dust, water and fungi. Dissimilar metals shall not be used in intimate contact without protection from galvanic corrosion. Asbestos, Cadmium and radioactive materials shall not be used in this item. Radioactive material is defined by Title 10, Code of Federal Regulations, Part 40, and other radioactive material in which is greater than 0.002 microcuries per gram or 0.01 microcuries total activity for the item.

3.3 Design and construction. The auger shall consist of a turntable-mounted power-unit superstructure mounted on a heavy-duty, truck-type carrier unit. The auger shall be furnished with a water pump for delivering water to the dug holes. The rotating superstructure shall consist of a turntable, mounting platform and operating machinery. All operating machinery shall be mounted on or attached to the platform (see 4.5.2).

3.3.1 Turntable. The turntable shall rotate at least 160 degrees (°) to the left and the right from the centerline of the rear of the truck chassis. A swing lock shall be provided to lock the rotating superstructure when in travel position. A brake shall be provided to hold the rotating superstructure in position at any point of the swing.

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3.3.2 Mounting platform. The mounting platform shall be of a size to support all the operating machinery and the operator. An expandable metal or raised pattern deck plate not less than 0.3 centimeter (cm) (1/8 inch (in.)) thick shall cover the operator's working area of the top surface of the platform. The deck plate shall be painted with a nonskid, abrasive-type, deck point. The platform shall extend and retract not less than a distance of 1.2 meter (m) (4 feet (ft.)).

3.3.3 Crown block. The derrick shall be fitted with a crown block assembly. Sheaves shall be furnished with provisions for preventing the wire rope from jumping off the sheaves grooves and shall be equipped with self-lubricating antifriction bearings. An eve-type anchor shall be attached to the underside of the crown block.

3.3.4 Derrick. The derrick shall be constructed of an electrically welded steel tubing frame consisting of four legs reinforced with diagonal and cross members. The derrick shall be rated by its manufacturer to sustain a gross axial capacity of 27 200 kilogram (kg) (60 000 pounds (lb.)) and shall be of sufficient height to accommodate 9 m (30 ft.) long sections of casing. The derrick shall be equipped with kelly holders to attach the kelly bar to the derrick for transit. The derrick shall be raised and lowered hydraulically by two cylinders not less than 15 cm (6 in.) in diameter. Each cylinder shall be provided with one or more pilot-operated check valves.

3.3.5 Kelly. The kelly shall be of the telescoping type of not less than 13.3 cm (5.25 in.) square or 15.2 cm (6 in.) in diameter. The inner kelly shall have an individual hoist operated by a hand lever and controlled by a brake foot pedal. A lock or brake shall be incorporated to prevent the outer and inner kelly bars from telescoping when maximum down pressure is applied to the auger bit at any point during the digging operation to the full length of the extended kelly. An instrument shall be furnished to determine the position of the kelly from vertical to 45° in increments of not more than 5° and shall be visible to the operators when seated in the operator's seat.

3.3.6 Rotary table. The rotary table shall transfer the rotation from the power source to the kelly bar. The rotary table shall be furnished with a replaceable kelly bushing or rollers. The rotary table shall have not less than six rotary speeds forward and one or more reverse speeds. The maximum torque available at the kelly shall not be less than 19 800 Newton•meter (N•m) (175 000 inch•pounds (in•lb)) at 25 revolutions per minute (rpm) of the kelly bar. The highest rotary speed imparted to the kelly shall not be less than 120 rpm.

3.3.7 Power feed and retracting mechanism. The power feed and retracting mechanism shall provide pressure to the kelly to increase or decrease the pressure on the auger bit. The power feed shall impart a maximum travel speed to the kelly of not less than 50 m (165 ft.) per minute in both up and down directions. Means shall be provided for applying not less than 13 600 kg (30 000 lb.) down pressure along the entire travel of the kelly without regard to the position of the kelly.

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3.3.8 Weight and dimensions. The weight and dimensions of the auger without attachments shall not exceed the following:

Weight (Gross Vehicle Weight (GVW))	27 200 kg	(60 000 lb.)
Width	244 cm	(96 in.)
Height	406 cm	(160 in.)
Length	1341 cm	(528 in.)

3.3.9 Angle of approach and departure. The angle of approach and departure shall be a minimum of 17° front and rear.

3.3.10 Lighting. The auger shall be furnished with a 24 volt lighting system consisting of service and blackout lighting to allow daylight performance to be maintained during night operation. The service lighting, including identification, side-marker and clearance lights, shall conform to the requirements of the Federal Motor Carrier Safety regulations (FMCSR) and the Federal Motor Vehicle Safety Standards (FMVSS) found in Part 49 of the Code of Federal Regulations. A rotating warning light mounted in such a manner as to insure visibility from above and all side shall be furnished with color of dome as specified (see 6.2).

3.3.11 Carrier electrical system. The carrier shall be equipped with a 24 volt vehicular electrical system.

3.3.12 Diesel engines. The engines, including all systems, filters, generators, batteries, air cleaners, housings, instrumentation, components, accessories and auxiliaries shall be as specified in the contract or order (see 6.2), except as otherwise specified herein. The power and speed rating of each engine shall be such that the operation of the auger under any of the conditions specified herein, shall not require horsepower in excess of the intermediate horsepower of the engines for the applicable operating speed as specified by the manufacturer. The engine speed shall be controlled by a throttle at speeds from the minimum design speed to the maximum design speed of the unit being driven as recommended by the carrier or auger manufacturer. Accessories furnished with each engine shall include but shall not be limited to: air induction system, complete cooling system, electrical system, fuel system, exhaust system and engine governing system.

3.3.12.1 Engine clutch. The carrier shall be furnished with a heavy-duty, double-disk, truck-type clutch. The clutch shall be rated by its manufacturer at not less than 160 percent of the rated engine torque and shall transmit the full engine torque smoothly without slippage, chatter or fierce engagement.

3.3.13 Steering. The carrier shall be equipped with an integral, hydraulic, power-steering system conforming to the following requirements:

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a. The pump, reservoir and filter shall be as recommended by the power-steering gear manufacturer. Each hydraulic reservoir shall be equipped with a dipstick or sight gauge to determine the level of fluid. A drain shall be located for complete drainage of the fluid into a container upon removal of the drain plug.
b. The operating fluid pressure range shall be from 6200 kiloPascals (kPa) (900 pounds per square inch (psi)) to 10 300 kPa (1500 psi), operate with hydraulic fluid and provide ample fluid flow for the performance specified.

3.3.14 Service airbrake system. The carrier shall be equipped with a waterproof, straight airbrake system that shall comply with the requirements of FMCSR and FMVSS, and all wheels shall be provided with service brakes. An emergency-parking brake system shall be provided and shall include an isolated reserve air tank. The system shall perform as an emergency brake conforming to the California Motor Vehicle Code and shall also perform as a parking brake conforming to FMCSR and FMVSS. The operation of the emergency-parking brake system, due to a service brake failure and when manually applied, shall safely stop the auger. The emergency-parking brake system shall provide not less than two times the number of applications required in the California Motor Vehicle Code, Section 26508, under all conditions specified herein. When the air pressure is below the minimum operating pressure, the system shall prevent the operator in the seat from releasing the emergency-parking brakes. Manual or automatic application of the emergency-parking brake system shall stop the auger on level concrete from 32 kilometers per hour (km/h) (20 miles per hour (mph)) in not more than 30 m (100 ft.). The emergency-parking brake system shall hold the auger on a 40 percent grade. Loss of any diaphragm or air line shall result in not more than 50 percent reduction in the maximum braking effort.

3.3.15 Operator's seat. The operator's seat dimensions and ranges of adjustment shall be in accordance with SAE J899. Removable, foam-rubber-padded, waterproof-covered, back and bottom cushions shall be furnished.

3.3.16 Controls. Hand and foot operated controls shall be located in accordance with SAE J898, to enable the operator to perform all the operations including leveling the auger, positioning the kelly, rotating the turntable, sliding the platform, digging the hole and the setting the casing.

3.3.17 Emergency starting and charging receptacle and cable. Slave receptacles shall be provided to facilitate electrical connections between both the carrier and rotating superstructure batteries and an external electrical power source for emergency starting of the engines and charging of the batteries. Receptacles shall be accessible to the slave plugs specified below. Unless otherwise specified (see 6.2), a 9 m (30 ft.) long cable conforming to part No. 7728530 of Drawing 7056674 shall be furnished to connect the slave receptacle on the rotating superstructure and the slave receptacle on the carrier to an external source or to each other. The cable shall be equipped at each end with a slave plug assembly.

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3.3.18 Carrier. The carrier shall be a heavy-framed, 4 axle supported, rubber-tired, heavy-duty, self-propelled vehicle upon which the rotating superstructure is mounted. The rotating superstructure shall be positioned in a fixed location on the carrier. Safety tread steel plate not less than 0.3 cm (0.125 in.) in thickness shall completely cover all openings between the carrier longitudinal frame members. This safety-tread plate shall be fastened to the frame members by bolts.

3.3.19 Wheel drive. The wheel drive shall be powered from the carrier engine through gear-driven axles. The drive shall be 8 by 4; four axles, two front (tandem) and two rear (tandem) driven.

3.3.20 Carrier exhaust system. The exhaust system shall be routed vertically. The stack shall terminate as a truck-type curved stack pipe with not less than a 37 mm (14.5 in.) radius curve undercut to protect the exhaust system from the weather. The stack shall be made of aluminized steel tubing. An expanded or pierced metal guard shall protect personnel from burns. The exhaust stack shall be braced and clamped to prevent damage by vibration and shall be equipped with a removable clamp and upper exhaust pipe with flanges held by corrosion-resistant steel bolts, washers and nuts to facilitate reducing the stack height to below the upper half of the cab. The muffler shall not be located in the removable portion of the exhaust stack.

3.3.21 Transmission. The transmission for the carrier shall consist of a main transmission and an auxiliary transmission. The main transmission shall provide not less than four forward speeds and not less than one reverse speed. The auxiliary transmission shall provide not less than three speeds.

3.3.22 Axles. All driving axles shall be either double reduction or planetary reduction and shall be full floating.

3.3.23 Suspension. Front spring and walking beam solid rear suspension shall be furnished.

3.3.24 Outriggers. The carrier shall have outriggers consisting of hydraulically operated sliding beams housed in a suitable metal box attached at right angles to the frame of the carrier. The outriggers shall be located fore and aft of the rear tandem wheels and on each side of the carrier. The outriggers shall be complete with not less than four hydraulic jacks and jack pads. Each jack pad shall have not less than 645 square centimeter (cm^2) (100 square inch (in^2.)) of area. With the auger on level ground the jacks shall be long enough to lift the rear tandem tires free of the ground a distance of not less than 10 cm (4 in.) without the use of blocks. Leveling jacks shall be individually hydraulically operated.

3.3.25 Towing attachments. Two towing eyes, shackles or hooks shall be mounted on the front of the carrier. When the towing attachments are mounted underneath the chassis, they shall be of the eye type and shall be not less than 7.6 cm (3 in.) in inside diameter. Each towing

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attachment on the front shall withstand a pull in the longitudinal direction equal to the maximum Gross Vehicle Weight (GVW) of the auger. The towing attachments shall be accessible for attaching a tow bar, chain or cable.

3.3.26 Lubrication. All surfaces requiring lubrication shall be provided with accessible lubrication fittings. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.3.27 Oil lubrication. Unless otherwise preserved for packaging (see 5.1), all components shall be filled to the operating level with oils designated for the use at 0 degrees Celsius (^{o}C) (2 degrees Fahrenheit (^{o}F)) temperature. Each lubricated component shall be tagged in a conspicuous place to indicate the temperature ranges and grades of oils used.

3.3.28 Accessories. Unless otherwise specified (see 6.2), the accessories specified in table I shall be furnished with each auger:

TABLE I. Accessories to be furnished.

No.	Accessories	Quantity
1.	Auger bits, all purpose type, minimum length 1.4 m (4.5 ft.), with hardface tungsten-carbide teeth (Williams 201 Jiffy type or equal and stinger(Williams HF type or equal), Williams type HC or equal.	
	a. Size 51 cm (20 in.)	1 ea.
	b. Size 61 cm (24 in.)	1 ea.
	c. Size 76 cm (30 in.) with adjustable reamer blades, drilled for 2.5 and 5 cm (1 and 2 in.) extensions on the radius	1 ea.
	d. Size 91 cm (36 in.)	1 ea.
2.	Bucket, bailing, 61 cm (42 in.) dia, 1.2 to 1.5 m (4 to 5 ft.) long	1 ea.
3.	Casing, welded construction, 0.6 cm (0.25 in.) wall thickness, 76 cm (30 in.) inside dia., 9 m (30 ft.) long, 0.6 x 15 cm (0.25 x 6 in.) reinforcing ring at one end with spinner and clevis holes.	54 m (180 ft)
4.	Sling, 2 legs, with 7.6 cm (3 in.) inside dia. ring and clevis on each leg, 1.2 m (4 ft.) minimum overall length	2 ea.
5.	Spinner, casing, universal, 61 to 91 cm (24 to 36 in.)	1 ea.
6.	Stingers, hardfaced tungsten-carbide, (Willaims HF type or equivalent)	3 ea.
7.	Teeth, hardfaced tungsten-carbide (Williams 201 Jiffy type or equivalent)	100 ea.

3.3.29 Fenders. Wheel housings and fenders or other equivalent protection shall be provided for all wheels to preclude throwing of debris. The rear fenders shall be constructed of sheet steel not less than 3 mm (0.11 in.) in thickness and where used as a walkway shall have a raised floortread pattern.

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3.3.30 Cab. The cab shall be of the tandem type with a seat for one passenger behind the driver. The outside width of the cab shall be not less than 81 cm (32 in.). The cab shall be offset to the left side of the chassis. The cab shall have two doors. A metal visor attached above the top of the windshield or a visor formed by extending the cab roof beyond the top of the windshield shall be furnished. Each control, switch, gauge and indicator light shall be labeled to indicate function and, where necessary, to indicate operation. An adjustable west-coast-type, outside, rearview mirror having an overall width of not less than 15 cm (6 in.) shall be placed on the driver's door of the carrier cab. The reflective surface shall be not less than 560 cm^2 (88 in^2.). The mirror shall be braced both top and bottom to prevent blurred vision because of vibration at any highway speed. A second, smaller, rearview mirror shall be placed to permit the driver to see the off-road side of the auger. The floor shall be covered with a removable neoprene floor mat. The windshield glass shall be laminated safety plate glass. The remainder of the cab glass shall be 0.6 cm (1/4 in.), tempered, safety plate glass. An adjustable, weather resistant, safety seat belt with a noncorrosive metal-to-metal, quick-release buckle shall be furnished for each seat and shall meet all requirements of SAE J1834. Anchorage shall be in accordance with SAE J383 for "Seat Belts Attached to the Seat Frame."

3.3.31 Heater and defroster. A fresh-air-type heater with defroster shall be furnished. The heater and defroster shall conform to all applicable requirements of the FMCSR and FMVSS.

3.3.32 Derrick rack. A rack located at the front of the carrier shall be provided for securing the derrick when in a travel position. The rack shall rigidly secure the derrick from swinging and striking the carrier cab. A wood filler strip shall be placed on top of the rack to prevent damage to the derrick during periods when the derrick may be rested on the rack for maintenance. The rack shall be reducible to a height not greater than the height of the cab by disassembly using bolts, star-type lockwashers and captive nuts welded to the framework. A support shall be provided for the front of the platform during transport.

3.3.33 Undercoating. The wheel wells, fenders and sheetmetal exposed to road splash shall be undercoated to the manufacturer's specifications.

3.3.34 Pocket. Two or more pockets shall be furnished on the sides of the carrier for transporting the 91 cm (36 in.) auger and spinner specified in 3.3.28.

3.3.35 Chock blocks. Two wheel chocks conforming to Drawing 8343584, with chains attached to the frame, shall be provided. The chain length shall permit chocks to be placed before and behind the auger's tires. Chock block cradles conforming to Drawing 8343583 shall be provided for securing the chocks while in transit.

3.3.36 V-belts. V-belts shall conform to the manufacturer's specification and SAE J636. All belts for multiple-belt drives shall be furnished as specified in the contract or order (see 6.2).

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3.3.37 Hose. All hoses shall be resistant to environmental and operating conditions. Metal fittings shall be protected from corrosion. Rubber components shall be in accordance with ASTM D380.

3.3.38 Safety standards and regulations. Each auger shall conform to all applicable requirements of the Federal Motor Carrier Safety Regulations (FMCSR) and the Federal Motor Vehicle Safety Standards found in Part 49 of the Code of Federal Regulations, except as specified.

3.3.39 Safety. All exposed parts which are subjected to high operating temperatures or which are energized electrically shall be insulated, enclosed or guarded. All moving parts, except the kelly bar, which are of such nature or so located as to be a hazard to operating or maintenance personnel and subject to breakage or entanglement with wire ropes shall be enclosed or guarded. Guards shall be provided to prevent dripping of grease or oil onto other components. Protective devices shall not impair the operating functions.

3.3.40 Maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair and replacement. Covers, plates and safety guards which must be removed for component adjustment, servicing, repair, replacement or maintenance shall be plainly matchmarked for ease of assembly and for adjustment. Aligning devices such as dowels and guides shall be used to aid reassemble. Items such as connections, plugs and valves shall be standardized types, and the numbers of different sizes shall be held to a minimum. Drain holes shall be furnished with facilities, where required, for draining to the ground or with space for draining into a 7.6 liter (L) (2 gallon) bucket.

3.3.41 Tires, rims and wheels. All units shall be equipped with suitable size wheels and tires, in accordance with the recommendations of the Tire and Rim Association, to withstand the maximum loads, stresses and travel speeds with adequate road clearance. A spare tire, tube and wheel (including spare-tire carrier) shall be furnished with each carrier and shall be located in an accessible location. Under normal highway travel conditions, the load on each tire shall not be greater than 2430 kg (5340 lb.).

3.4 Performance.

3.4.1 Vertical hole digging. The auger shall be capable to dig 91 cm (36 in.) holes to a depth of 18 m (60 ft.) and be capable to seat 18 m (60 ft.) of 76 cm (30 in.) diameter, 9 m (30 ft.) long casing (see 4.5.4.1).

3.4.2 Angular hole digging. The auger shall be capable to dig holes from vertical to 45° for the entire length of the extendible kelly. A gauge shall be provided with increments of not more than 5 degrees and shall be visible to the operator when seated in the operator's seat (see 4.5.4.2).

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3.4.3 Penetration rate. The auger shall have a penetration rate of not less than 61 cm (24 in.) in soil and not less than 30 cm (12 in.) in rock (see 4.5.4.3).

3.4.4 Accessory handling. The auger shall be equipped with a service hoist for removal of auger bits from the storage pocket and assist in the installation of the bit (see 4.5.4.4).

3.4.5 Pump. The auger shall be equipped with a pump capable of delivering water at 190 L/min (liters per minute) (50 gallons per minute(gpm)) with 24 m (80 ft.) head to the hole being drilled (see 4.5.4.5).

3.4.6 Service winch and line. A two-speed service winch shall be furnished with a baredrum-line pull of not less than 7300 kg (16 000 lb.) at low speed and 3600 kg (8000 lb) at high speed. The maximum bare drum-line speed shall be not less than 27 m (90 ft) per minute at low speed and 54 m (180 ft.) per minute at high speed. The brake lever shall be of the positive-locking type. The service winch drum shall have a spooling capacity of not less than 67 m (200 ft.) of 1.6 cm (5/8 in.) diameter wire rope with a nominal strength of 13 000 kg (28 600 lb.). The free end of the rope shall be fitted with a safety clevis (see 4.5.4.6). All components in the winch system shall have at least a safety factor of two.

3.4.7 Gradeability and travel speed. The auger shall be able to transverse a 1 percent grade at speeds of not less than 56 km/h (35 mph). The auger shall also be able to transverse a 40 percent grade (see 4.5.4.7).

3.4.8 Road test. The auger shall be able to travel secondary roads at speeds of not less than 16 km/h (10 mph) (see 4.5.4.8).

3.4.9 Rail transport. The auger shall be sectionalized to allow for rail transport (see 4.5.4.9).

3.4.10 Kelly bar. The auger shall have a maximum torque capacity of 20 000 N•m (175 000 in•lb.) at speeds of 25 rpm, maximum speeds of 50 m (165 ft.) per minute and a minimum down pressure of 13 000 kg (30 000 lb.) (see 4.5.4.10).

3.4.11 Emergency-parking brake system. The auger shall be equipped with an emergency-parking brake system that will stop the auger in event of the failure of the main brake system and hold the auger on 40 degree grades (see 4.5.4.11).

3.4.12 Operation. The auger shall allow for continuous operation as specified herein (see 4.5.4.12).

3.4.13 Lifting and tiedown attachments. The complete auger shall be equipped with lifting and tiedown attachments. Each disassembled component shall also be equipped with tiedown attachments. The attachments shall be located so that clearance is maintained between the slings and all exterior parts of the equipment to be lifted or tied down. Lifting attachments

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may also be used as tiedown devices when such attachments meet the requirements for tiedown devices. Lifting and tiedown devices shall be marked to indicate their purpose (see 4.5.4.13).

3.5 Marking. The auger shall be marked with the National Stock Number (NSN) and the manufacturer's name and part number. Markings shall be permanent and legible (see 4.5.2).

3.6 Instruction plates. Each control shall be marked to identify its function and to include instructions, warnings, and cautions where damage to the auger or danger to personnel could result due to incorrect settings during starting, operation, or shutdown of the auger (see 4.5.2).

3.7 Treatment and painting. The treatment and painting of the auger shall be to the manufacturer's specifications (see 4.5.2).

3.8 Technical publications to accompany auger. Such technical publications as are specified shall be furnished (see 4.5.2 and 6.2).

3.9 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished (see 4.5.2 and 6.2).

3.10 Items to be furnished with the auger. No shipment of augers shall be made unless repair parts, maintenance tools, and accessories are included with the shipment or unless approval for shipment of the augers without such items has been received from the contracting officer (see 4.5.2).

3.11 Toolbox. A toolbox shall be furnished and mounted on the auger. The toolbox shall have not less than 33 cu. cm (2 cu. ft.) of storage space. Material used shall be steel not less than 0.2 cm (0.071 in.) thick. Provisions shall be incorporated for locking the toolbox. Drainage shall be provided as mounted (see 4.5.2).

3.12 Workmanship. The auger shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair the operation or serviceability of the auger (see 4.5.2).

3.13 Noise limits. Noise levels shall not exceed 85 dB(A) at operating and occasionally occupied positions during typical load operations. If it can be shown that 85 dB(A) is beyond the state of the art for this equipment, than a permanent hearing protection warning decal shall be placed on the equipment stating that hearing protection is required within (specify) feet. the distance specified shall be the greatest distance at which 85 dB(A) is met. The decal shall be readable to all personnel exposed to the hazard. Warnings shall also be added to applicable manuals requiring hearing protection to be worn within the specific distance (see 4.5.2).

4. VERIFICATION

4.1 Classification of inspection. The inspection conditions specified herein are classified as follows:

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a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on preproduction or initial production samples as specified when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.4 (see table II) and the tests of 4.5.4.1 through 4.5.4.13 (see table III).

4.2.1 Preproduction inspection. When specified (see 6.2), first article preproduction inspection shall be performed on two augers. This inspection shall include the examinations of 4.4 (see table II) and the tests of 4.5.4.1 through 4.5.4.13 (see table III).

4.2.2 Initial production inspection. Unless otherwise specified (see 6.2), first article preproduction inspection shall be performed on two augers. This examination shall include the examinations 4.4 (see table II) and the tests of 4.5.4.1 through 4.5.4.13 (see table III).

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.4 and the tests of 4.5.4.1, 4.5.4.2, 4.5.4.3, 4.5.4.5 and 4.5.4.10 through 4.5.4.12 (see table III).

4.4 Examination.

4.4.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ANSI/ASQC Z1.4. Any redesign or modification of the contractor's standard to comply with specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements as listed in table II. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Method of inspection.

4.5.1 Materials. Conformance to 3.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports and rating data.

4.5.2 Defects. Conformance to 3.3 through 3.3.41 and 3.5 through 3.13 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile or by measurement with SIE.

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4.5.3 Test conditions. Unless otherwise specified (see 6.2), test conditions shall be as specified herein.

4.5.4 Performance.

4.5.4.1 Vertical hole digging. To determine conformance to 3.4.1, this test shall be conducted at an actual or simulated altitude 1500 m (5000 ft.). If the option is taken to simulate altitude conditions for test purposes, such simulation shall be accomplished in the following manner. In the vicinity of the engine, mount a container of sufficient capacity and inlet orifice size to assure a steady air pressure to the engine equal to 63.2 cm (24.89 in.) of mercury for 1500 m (5000 ft.) Connect the outlet of the container to the air cleaner so that air is inducted into the engine in the normal manner through the air cleaner filter. Determine the intake air pressure with a U-tube manometer tapped into the container at a point where static air pressure is maintained in the container. The intake and exhaust systems shall be complete and unaltered, except that the carburetor may be adjusted for optimum performance. Set up the auger at the digging site. Dig three holes using a 91 cm (36 in.) bit to a minimum depth of 18 m (60 ft.) in the soil and rock and perform the following operations:

a. Set 18 m (60 ft.) of 76 cm (30 in.) diameter casing in one of the holes (sections of casing shall be 9 m (30 ft.) long).
b. Dig one hole at night using the lights mounted on the auger as the only illumination.
c. Maintain a digging log and indicate the following:
    1. Formation encountered and at what depth.
    2. Digging time.
    3. Shutdown and reasons (including service and maintenance).

Submit to the contracting officer a certified reproducible copy of the log. Inability to complete the test, reduced digging rate at night, digging log incomplete, any permanent deformation or breakage to components of the auger (bit teeth excluded), or not disposing of the soil shall constitute failure of this test.

4.5.4.2 Angular hole digging. To determine conformance to 3.4.2, this test shall be conducted by digging in soil and rock. Using the 91 cm (36 in.) auger bit, dig one hole 30° from vertical and a second hole 45° from vertical to the fully extended length of the telescopic kelly. Accuracy of the angle gauge shall be determined by comparing to actual kelly bar angle. Inability to complete the test, inability to complete the test, inability of the operator to read the angle gauge and determine within 2 degree tolerance the prescribed angle, or any permanent deformation or breakage to components of the auger (bit teeth excluded) shall constitute failure of this test.

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4.5.4.3 Penetration rate. To determine conformance to 3.4.3, this test shall be conducted with the 91 cm (36 in.) auger resting on the bottom of a hole not less 0.9 cm (3 ft.) deep. The kelly bar shall be marked to provide a point to measure kelly bar travel. A record of elapsed time shall be taken from the moment the operator initially engages the kelly bar down-travel control to the moment of final release. The elapsed time of the test shall not be less than 1 minute. Measure the travel of the kelly bar. Calculate the penetration rate in both soil and rock. A penetration rate of less than 61 cm/min (2 ft./min) in soil or a penetration rate less than 30.5 cm/min (1 ft./min) in rock shall constitute failure of this test.

4.5.4.4 Accessory handling. To determine conformance to 3.4.4, this test shall be performed by two men. A 91 cm (36 in.) auger bit shall be stored for transport in a storage pocket on the auger. Using the auger service hoist, pick up the 91 cm (36 in.) auger bit from the storage pocket and deposit it on the ground within 1.5 m (5 ft.) of the carrier. Insert the kelly bar stub in the bit socket and insert locking pin. Inability to complete the test or requiring additional personnel or equipment shall constitute failure of this test.

4.5.4.5 Pump. To determine conformance to 3.4.5, correct all data to standard conditions. Suction pipe and fittings shall be 5 cm (2 in.) nominal size.

a. Subject the pump to an internal hydrostatic pressure of 1-1/2 times the shutoff head pressure for not less than 1 minute.
b. After initially filling the pump, operate the pump at static suction lifts equivalent to 3 m (10 ft.).
c. Operate the pump at continuous duty rpm and static suction lift equivalent to 3 m (10 ft.); continue testing the pump for conformance to the head capacity of 190 L/min (50 gpm) at 24 m (80 ft.) head or greater shall constitute failure of this test.

4.5.4.6 Hoist drumline pull and speed test. To determine conformance to 3.4.6, apply a 7300 kg (16 000 lb.) load to the service hoist line and have not more than two wraps of rope around the hoist drum. Engage the hoist at low speed and lift the load. Calculate the bare drum line speed. Repeat the procedure applying an 3650 kg (8000 lb.) load and employing the high speed of the hoist. Inability to lift the loads, bare drum line speeds less than specified in 3.4.6 or permanent deformation or breakage to any component of the auger shall constitute of this test.

4.5.4.7 Gradeability and travel speed. To determine conformance to 3.4.7, this test will be conducted on a hard surface roadway with uniform grade. From a standing start, travel the auger up the 40 percent grade for a distance of not less than 9 m (30 ft.) Travel the auger at 56 km/h (35 mph) up the 1 percent grade for not less than 300 m (1000 ft.). Inability to transverse the 40 percent grade for a distance of 9 m (30 ft.) or to maintain a speed of at least 56 km/h (35 mph) over the 1 percent grade for a distance of 300 m (1000 ft) shall constitute failure of this test.

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4.5.4.8 Road test. To determine conformance to 3.4.8, drive the auger, loaded with accessories, tools and equipment necessary to dig a 91 cm (36 in.) hole, a distance of not less than 370 km (200 miles) over secondary roads. Turn on all lights controlled by the carrier. During the travel over this 370 km (200 miles), shift the transmission both up and down through each transmission speed not less than 50 times and come to a complete stop after each shift cycle. A travel speed of not less than 16 km/h (10 mph) shall be maintained throughout this test. Inability to complete the test, damage or malfunction to any component or part, leakage or overheating of any part shall constitute failure of this test.

4.5.4.9 Rail transport. To determine conformance to 3.4.9, sectionalize the auger for rail transport and reassemble.

4.5.4.10 Kelly bar. To determine conformance to 3.4.10, measure the maximum torque available, the maximum travel speed in both directions and the maximum downward pressure at 1/3, 2/3 and 7/8 extended position. Inability to develop a torque of 19 800 N-m (75 000 psi) at 25 rpm, maximum speed of 50 meters per minute (m/min) (165 feet per minute (fpm)) in both directions and 13 600 kPa (30 000 lb) down pressure shall constitute failure of this test.

4.5.4.11 Emergency-parking brake system. To determine conformance to 3.4.11, drive the auger forward over a dry level concrete surface with a grade not exceeding 1 percent. When the auger reaches a speed of not less than 32 km/h (20 mph), manually actuate the emergency-parking brake system. Measure and record the distance between the point of actuation and the point of stoppage. Drive the auger onto the test course and stop. Apply the emergency-parking brakes by exhausting service reservoir air to the point that the emergency system is automatically actuated.

4.5.4.12 Operation. To determine conformance to 3.4.12, drive the auger a distance of not less than 16 km (10 miles). Shift the transmission both up and down through each transmission speed not less than five times, coming to a complete stop after each cycle. Set up the auger for hole digging operation. Dig 91 cm (36 in.) holes in earth to any desirable depths, operating for not less than 2 hours. All major components and all controls shall be operated to determine performance. Any malfunction, breakage or permanent deformation shall constitute failure of this test.

4.5.4.13 Lifting and tiedown attachments. To determine conformance to 3.4.13, lift the disassembled sections of the auger and determine the normal load on each attachment. Apply 2-1/2 times the normal load to each attachment in the direction in which a sling normally applies the load. Repeat the procedure on the assembled auger. Then anchor each disassembled section of the auger to a flatcar using the tiedown attachments and no blocking, bracing, or cribbing. Hump the loaded flatcar at a speed of 11 km/h (7 mph) against five boxcars with their airbrakes set. Repeat the humping with the opposite end of the section in the direction of the five cars.

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Repeat the procedure on each assembled auger. Weld failure or permanent deformation of the attachments or any part of the auger shall constitute failure of this test.

TABLE II. Classification of defects.

Category	Defect	Method of Examination
101	Material not as specified (see 3.2)	Visual
102	Turntable not as specified (see 3.3.1)	Visual
103	Mounting platform not as specified (see 3.3.2)	Visual
104	Crown block not as specified (see 3.3.3)	Visual
105	Derrick not as specified (see 3.3.4)	Visual
106	Kelly not as specified (see 3.3.5)	Visual
107	Rotary table not as specified (see 3.3.6)	Visual
108	Power feed and retracting mechanism not as specified (see 3.3.7)	Visual
109	Weight and dimensions not as specified (see 3.3.8)	SIE 1/
110	Angle of approach and departure not as specified (see 3.3.9)	SIE
111	Lighting not as specified (see 3.3.10)	Visual
112	Carrier electrical system not as specified (see 3.3.11)	Visual
113	Diesel engines not as specified (see 3.3.12)	Visual
114	Engine clutch not as specified (see 3.3.12.1)	Visual
115	Steering not as specified (see 3.3.13)	Visual
116	Service airbrake system not as specified (see 3.3.14)	Visual
117	Operator's seat not as specified (see 3.3.15)	Visual
118	Controls not as specified (see 3.3.16)	Visual
119	Emergency starting/charging receptacle/cable not as specified (see 3.3.17)	Visual
120	Carrier not as specified (see 3.3.18)	Visual
121	Wheel drive not as specified (see 3.3.19)	Visual
122	Carrier exhaust system not as specified (see 3.3.20)	Visual
123	Transmission not as specified (see 3.3.21)	Visual
124	Axles not as specified (see 3.3.22)	Visual
125	Suspension not as specified (see 3.3.23)	Visual
126	Outriggers not as specified (see 3.3.24)	Visual
127	Towing attachments not as specified (see 3.3.25)	Visual
128	Lubrication not as specified (see 3.3.26)	Visual
129	Oil lubrication not as specified (see 3.3.27)	Visual
130	Accessories not furnished as specified (see 3.3.28)	Visual
131	Fenders not as specified (see 3.3.29)	Visual
132	Cab not as specified (see 3.3.30)	Visual
133	Heater and defroster not as specified (see 3.3.31)	Visual
134	Derrick rack not as specified (see 3.3.32)	Visual
135	Undercoating not as specified (see 3.3.33)	Visual
136	Pockets not as specified (see 3.3.34)	Visual
137	Chock blocks not as specified (see 3.3.35)	Visual
138	V-belts not as specified (see 3.3.36)	Visual

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TABLE II. Classification of defects - continued

Category	Defect	Method of Examination
139	Hoses not as specified (see 3.3.37)	Visual
140	Auger does not conform to FMCSR and FMVSS except as specified (see 3.3.38)	Visual
141	Safety devices or guards missing (see 3.3.39)	Visual
142	Assemblies not accessible as specified (see 3.3.40)	Visual
143	Tires, rims and wheels not as specified (see 3.3.41)	Visual
144	Marking not as specified (see 3.5)	Visual
145	Instruction plates not as specified (see 3.6)	Visual
146	Treatment and painting not as specified (see 3.7)	Visual
147	Technical publications not as specified (see 3.8)	Visual
148	Repair parts and maintenance tools not as specified (see 3.9)	Visual
149	Items to be furnished with auger not furnished (see 3.10)	Visual
150	Toolbox not as specified (see 3.11)	Visual
151	Workmanship not as specified (see 3.12)	Visual

1/ = Standard Inspection Equipment.

TABLE III. Classification of inspections

Title	Requirement	Inspection	First article	Quality conformance test
Vertical hole digging	3.4.1	4.5.4.1	X	X
Angular hole digging	3.4.2	4.5.4.2	X	X
Penetration rate	3.4.3	4.5.4.3	X	X
Accessory handling	3.4.4	4.5.4.4	X	
Pump	3.4.5	4.5.4.5	X	X
Service winch and line	3.4.6	4.5.4.6	X	
Gradeability and travel speed	3.4.7	4.5.4.7	X	
Road test	3.4.8	4.5.4.8	X	
Rail transport	3.4.9	4.5.4.9	X	
Kelly bar	3.4.10	4.5.4.10	X	X
Emergency-parking brake system	3.4.11	4.5.4.11	X	X
Operation	3.4.12	4.5.4.12	X	X
Lifting and tiedown attachments	3.4.13	4.5.4.13	X	

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or defense Agency's automated packaging files, CD-ROM products or by contracting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The auger furnished under this specification, is intended for military application and general construction use for digging holes in soil and soft rock.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).
c. If first article samples are not required (see 3.1).
d. Color of the dome required (see 3.3.10).
e. Engine, components and accessories (see 3.3.12).
f. When emergency cable is not required (see 3.3.17).
g. Accessories required (see 3.3.28).
h. V-belts required (see 3.3.36).
i. Technical publications required (see 3.8).
j. Repair parts and maintenance tools required (see 3.9).
k. If preproduction inspection is required (see 4.2.1).
l. If initial production inspection is not required (see 4.2.2).
m. Test conditions if other than as specified (see 4.5.3).
n. Packaging requirements (see 5.1).

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6.3 Subject term (keyword) listing.

Casings
Diesel engine
Digging
Holes

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 3820-0014)

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